Mathematics of the DEMReg approach

Iain Hannah

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For more detail, see Hannah & Kontar (2012, 2013); Kontar et al. (2004); Hansen (1992).

Regularization for the DEM problem

The observed data value g_i for channel or filter i is related to the underlying differential emission measure (DEM), $\xi(T) = n^2\,\mathrm{d}h/\mathrm{d}T$ [cm^{-5} K^{-1}], and the temperature response $K_i(T)$ by

$$g_i = \int K_i(T)\,\xi(T)\,\mathrm{d}T. \tag{1}$$

After discretising temperature into bins T_j, this can be written as a system of linear equations,

$$g_i = \sum_{j=1}^{N} K_{ij}\,\xi_j, \tag{2}$$

where $i = 1, \ldots, M$ and $j = 1, \ldots, N$. Typically $M < N$; for example, with six SDO/AIA channels and 30 temperature bins we would have $M = 6$ and $N = 30$.

To solve Eqn. 1 and recover the DEM, we seek

$$\boldsymbol{\xi}_{\mathrm{LS}} = \arg\min_{\boldsymbol{\xi}} \left\|\mathbf{K}\boldsymbol{\xi} - \mathbf{g}\right\|^2, \tag{3}$$

where $\|\mathbf{x}\|^2 = \mathbf{x}^T\mathbf{x} = \sum_{i=1}^{M} x_i^2$, the L_2 (Euclidean) norm.

Equation 1 defines an ill-posed inverse problem, so additional linear constraints are introduced to suppress noise amplification. A common choice is zero-order regularization, which selects the smallest-norm solution from the infinitely many admissible solutions. Instead of solving Eqn. 3, we therefore solve

$$\arg\min_{\boldsymbol{\xi}} \left\|\mathbf{K}\boldsymbol{\xi} - \mathbf{g}\right\|^2 \qquad \text{subject to} \qquad \left\|\mathbf{L}(\boldsymbol{\xi} - \boldsymbol{\xi}_0)\right\|^2 \le \mathrm{const.} \tag{4}$$

This can be solved using Lagrange multipliers, giving

$$\boldsymbol{\xi}_\lambda = \arg\min_{\boldsymbol{\xi}} \left(\left\|\mathbf{K}\boldsymbol{\xi} - \mathbf{g}\right\|^2 + \lambda \left\|\mathbf{L}(\boldsymbol{\xi} - \boldsymbol{\xi}_0)\right\|^2\right), \tag{5}$$

where $\mathbf{L}$ is the constraint matrix, λ is the regularization parameter, and $\boldsymbol{\xi}_0$ is an optional initial guess.

GSVD to solve the regularized problem

A solution to Eqn. 5 can be found via the GSVD as a function of λ (Hansen 1992). For $\mathbf{K} \in \mathbb{R}^{M\times N}$ and $\mathbf{L} \in \mathbb{R}^{N\times N}$, the GSVD yields singular values α_k and β_k (with $\phi_k = \alpha_k/\beta_k$) and singular vectors $\mathbf{u}_k$, $\mathbf{v}_k$, and $\mathbf{w}_k$ for $k = 1, \ldots, N$, satisfying $\alpha_k^2 + \beta_k^2 = 1$, $\mathbf{U}^T\mathbf{K}\mathbf{W} = \mathrm{diag}(\boldsymbol{\alpha})$, and $\mathbf{V}^T\mathbf{L}\mathbf{W} = \mathrm{diag}(\boldsymbol{\beta})$. Because in our case $M \le N$, only the first M singular values α_k are non-zero.

The solution is then, as given in Eqn. 18 of Hansen (1992),

$$\boldsymbol{\xi}_\lambda = \sum_{k=1}^{M} \left(f_k \frac{\mathbf{g}\cdot\mathbf{u}_k}{\alpha_k} + (1-f_k)\,\mathbf{w}_k^{-1}\boldsymbol{\xi}_0 \right) \mathbf{w}_k, \tag{6}$$

where the filter factors f_k are given by

$$f_k = \frac{\phi_k^2}{\phi_k^2+\lambda} = \frac{\alpha_k^2}{\alpha_k^2+\lambda\beta_k^2}, \qquad 1-f_k = \frac{\lambda\beta_k^2}{\alpha_k^2+\lambda\beta_k^2}. \tag{7}$$

In the code, Eqn. 6 is implemented as

$$\boldsymbol{\xi}_\lambda = \sum_{k=1}^{M} \frac{\left(\alpha_k(\mathbf{g}\cdot\mathbf{u}_k) + \lambda\beta_k^2\mathbf{w}_k^{-1}\boldsymbol{\xi}_0\right)\mathbf{w}_k}{\alpha_k^2+\lambda\beta_k^2}. \tag{8}$$

This is how it is done in the original IDL code[1] that accompanied Hannah & Kontar (2012). In the map versions (Hannah & Kontar 2013) of the IDL[2] and Python[3] code, no initial guess is used and $\boldsymbol{\xi}_\lambda$ is calculated slightly differently, with the regularized inverse of $\mathbf{K}$, $\mathbf{R}_\lambda \simeq \mathbf{K}^\dagger$, computed directly as

$$\mathbf{R}_\lambda = \sum_{k=1}^{M} \frac{\alpha_k\left(\mathbf{u}_k\mathbf{w}_k\right)}{\alpha_k^2+\lambda\beta_k^2}. \tag{9}$$

The DEM solution is then found from $\boldsymbol{\xi}_\lambda = \mathbf{R}_\lambda\mathbf{g}$. This form is used because $\mathbf{R}_\lambda$ is also needed to estimate the temperature resolution[4] of the DEM solution, so it does not need to be computed separately in the map versions, where computational speed is more important.

Note that:

- In Eqn. 6 of Hannah & Kontar (2012), the $\boldsymbol{\xi}_0$ term is not quite correct, although the code is.
- In Eqn. 24 of Kontar et al. (2004), no initial guess is shown, although one is included in the code[5].

[1] idl_org/dem_inv_reg_solution.pro
[2] idl/demmap_pos.pro
[3] python/demmap_pos.py
[4] idl_org/dem_inv_reg_resolution.pro
[5] ssw/packages/xray/idl/inversion/inv_reg_solution.pro

- In Eqn. 15 of Hansen (1992), the regularization parameter is written as λ^2 rather than λ, as used here.

To obtain a useful solution, we still need to choose suitable values of $\mathbf{L}$ and λ.

Choosing constraint matrix L

In most applications we consider zero-order regularization, for which $\mathbf{L} \propto \mathbf{I}$ and therefore has no off-diagonal terms. The diagonal elements of $\mathbf{L}$ are usually normalized by some scale, typically either the temperature-bin width, $\mathbf{L}_{jj} = 1/\mathrm{d}\log T_j$, or, when appropriate, an expected form for the final solution (which is not the same as the initial guess $\boldsymbol{\xi}_0$). One example is the minimum of the EM loci curves, $\boldsymbol{\xi}_{\mathrm{mlc}}$, taken as a smoothed and normalized version of the lowest emission measure at each temperature under an isothermal assumption; in that case $\mathbf{L}_{jj}$ is based on the minimum of g_i/K_{ij} over all i.

Choosing regularization parameter λ

The regularization parameter λ is chosen so that the solution $\boldsymbol{\xi}_\lambda$ from Eqn. 6 satisfies an additional criterion. The standard choice is Morozov's discrepancy principle, which effectively controls the χ^2 of the solution in data space. If the observations have associated uncertainties δg_i, then the preferred value of λ is the one that minimizes

$$\|\mathbf{K}\boldsymbol{\xi}_\lambda - \mathbf{g}\|^2 - \rho \, \|\boldsymbol{\delta g}\|^2 . \tag{10}$$

where ρ is a regularization-tuning parameter; by default we aim for $\rho = 1$.

From Eqn. 20 of Hansen (1992), the term $\|\mathbf{K}\boldsymbol{\xi}_\lambda - \mathbf{g}\|^2$ can be calculated directly from the GSVD products, namely

$$\|\mathbf{K}\boldsymbol{\xi}_\lambda - \mathbf{g}\|^2 = \sum_{k=1}^{M} (1 - f_k)^2 \left(\mathbf{g} \cdot \mathbf{u}_k - \alpha_k \mathbf{w}_k^{-1} \boldsymbol{\xi}_0\right)^2 . \tag{11}$$

In the code, Eqn. 11 is implemented as

$$\|\mathbf{K}\boldsymbol{\xi}_\lambda - \mathbf{g}\|^2 = \sum_{k=1}^{M} \left(\frac{\lambda\beta_k^2 \left(\mathbf{g} \cdot \mathbf{u}_k - \alpha_k \mathbf{w}_k^{-1} \boldsymbol{\xi}_0\right)}{\alpha_k^2 + \lambda\beta_k^2}\right)^2 . \tag{12}$$

This is how it is done in the original IDL code[6]. In the map versions of the IDL[7] and Python[8] code, Eqn. 12 should have used only $(\mathbf{g} \cdot \mathbf{u}_k)$ in the numerator because no initial guess, i.e. no $\boldsymbol{\xi}_0$, is used. However, $(\mathbf{g} \cdot \mathbf{u}_k - \alpha_k)$ was used instead, which may have affected the determination of the optimal λ. Crucially, the calculation of $\boldsymbol{\xi}_\lambda$ itself was correct in all versions. This has now been corrected in the map versions of the code.

[6] idl_org/dem_inv_reg_parameter.pro
[7] idl/dem_inv_reg_parameter_map.pro
[8] python/dem_reg_map.py

In all versions of the code, Eqn. 12 is evaluated over a range of λ values, with the range set by scaling the minimum and maximum of the ϕ_k values. The left-hand side of Eqn. 10 is then computed for each case, and the λ giving the smallest value is selected.

Using Eqn. 10 should provide the mathematically optimal solution, but not necessarily the most physical one, because it can return a $\boldsymbol{\xi}_\lambda$ with negative components. An additional positivity constraint can therefore be imposed, requiring the chosen λ to satisfy Eqn. 10 while also giving $\boldsymbol{\xi}_\lambda \geq 0$ component-wise. In the original IDL code[9], this is achieved by evaluating both Eqn. 12 and Eqn. 8 for a range of λ values and selecting the one that satisfies both criteria. In the map code (IDL[10] and Python[11]), Eqn. 12 was evaluated for a smaller number of λ samples and, if $\boldsymbol{\xi}_\lambda$ contained negative components, ρ was increased iteratively and Eqn. 12 recalculated until $\boldsymbol{\xi}_\lambda \geq 0$ or a maximum number of iterations was reached. This was faster than evaluating both Eqn. 12 and Eqn. 8 over a much larger λ sample, as in the original approach. A possible optimization of the original method would be to first identify a subset of λ values satisfying one criterion and then apply the second criterion only to that smaller subset; for example, first find the values that give $\boldsymbol{\xi}_\lambda \geq 0$ and then evaluate Eqn. 12 only on that subset.

Errors on the DEM solution

For more details on how these are calculated see Hannah & Kontar (2012, 2013) and the code. But in summary:

- The vertical errors on the DEM solution are obtained from linear propagation of the uncertainties in the input data.
- The horizontal errors on the DEM solution correspond to the temperature resolution and are estimated from the spread of the off-diagonal terms in $\mathbf{KR}_\lambda$. The better the solution $\boldsymbol{\xi}_\lambda = \mathbf{R}_\lambda \mathbf{g}$, the closer $\mathbf{KR}_\lambda$ is to $\mathbf{I}$.

Temperature response functions

The temperature response functions are given by

$$K_i(T_j) = \int_0^\infty G(\lambda, T_j)\, R_i(\lambda)\, \mathrm{d}\lambda, \tag{13}$$

where $G(\lambda, T)$ is the plasma emissivity in the solar atmosphere (the contribution function), derived from an appropriate solar model such as CHIANTI, and $R_i(\lambda)$ is the wavelength response of the i^{th} channel. These temperature response functions are usually precomputed and provided by the instrument teams, for example for SDO/AIA (Boerner et al. 2014). With calibrated EUV spectral-line data, however, the contribution functions $G_i(T_j)$ are used directly and can be calculated with CHIANTI's gofnt[12] function.

[9] idl_org/dem_inv_reg_parameter_pos.pro
[10] idl/demmap_pos.pro
[11] python/demmap_pos.py
[12] i.e. SSW version gofnt.pro

References

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